Food Delivery Order Performance Analysis

Data Visualization Report

Group C - DataVerse

Repository: Ahana9647/Food_Delivery_Project

Generated automatically on 10 October 2026

Records analysed: 50,000 orders, 17 original fields

1. Purpose and Method

This report presents the visual analysis of the cleaned food delivery dataset (*data/processed/cleaned_food_delivery_orders.csv*). All charts were created in Python using Pandas, Matplotlib and Seaborn and this PDF was generated automatically by *src/visualization.py*. The same summary tables are saved in *outputs/tables/* and loaded into Power BI for the interactive dashboard.

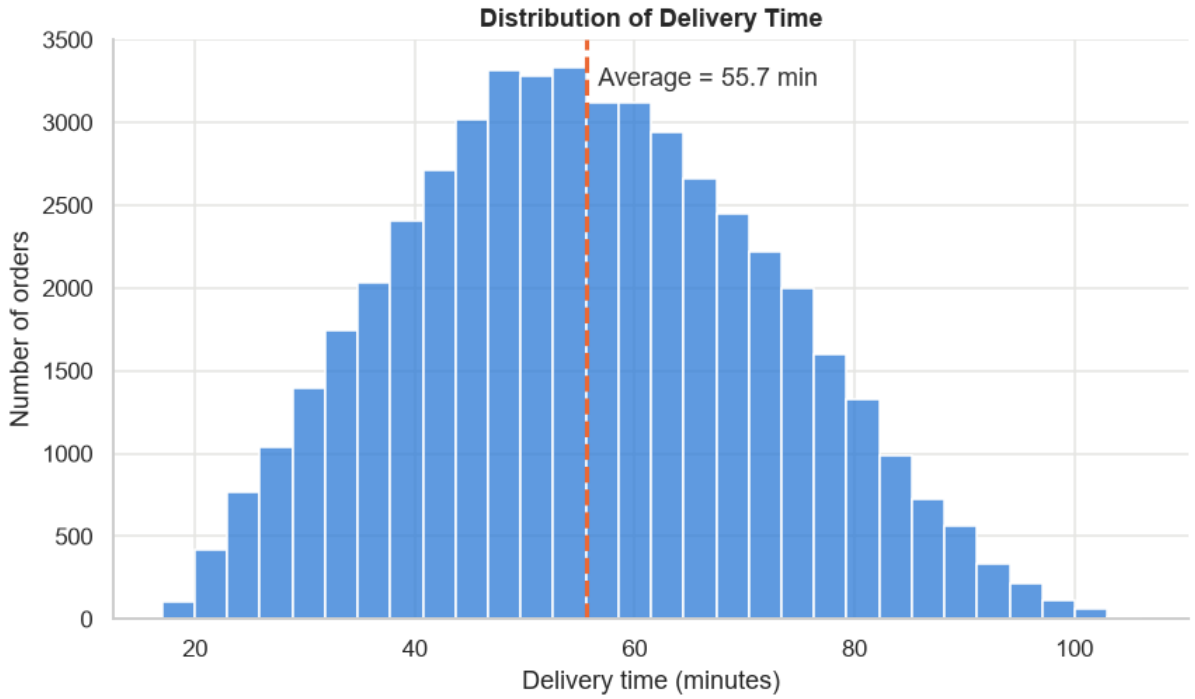
Scope of delivery-performance charts: all orders in the cleaned file.

2. Key Performance Indicators

KPI	Value
Total orders	50,000
Delivered orders	45,934
Cancelled orders	4,066
Average delivery time (min)	55.67
On-time delivery (%)	70.18
Late orders	14,911
Average rating	4.17
Average distance (km)	9.53
Average preparation time (min)	29.49
Average order value	1,401.50
Average delivery fee	36.00

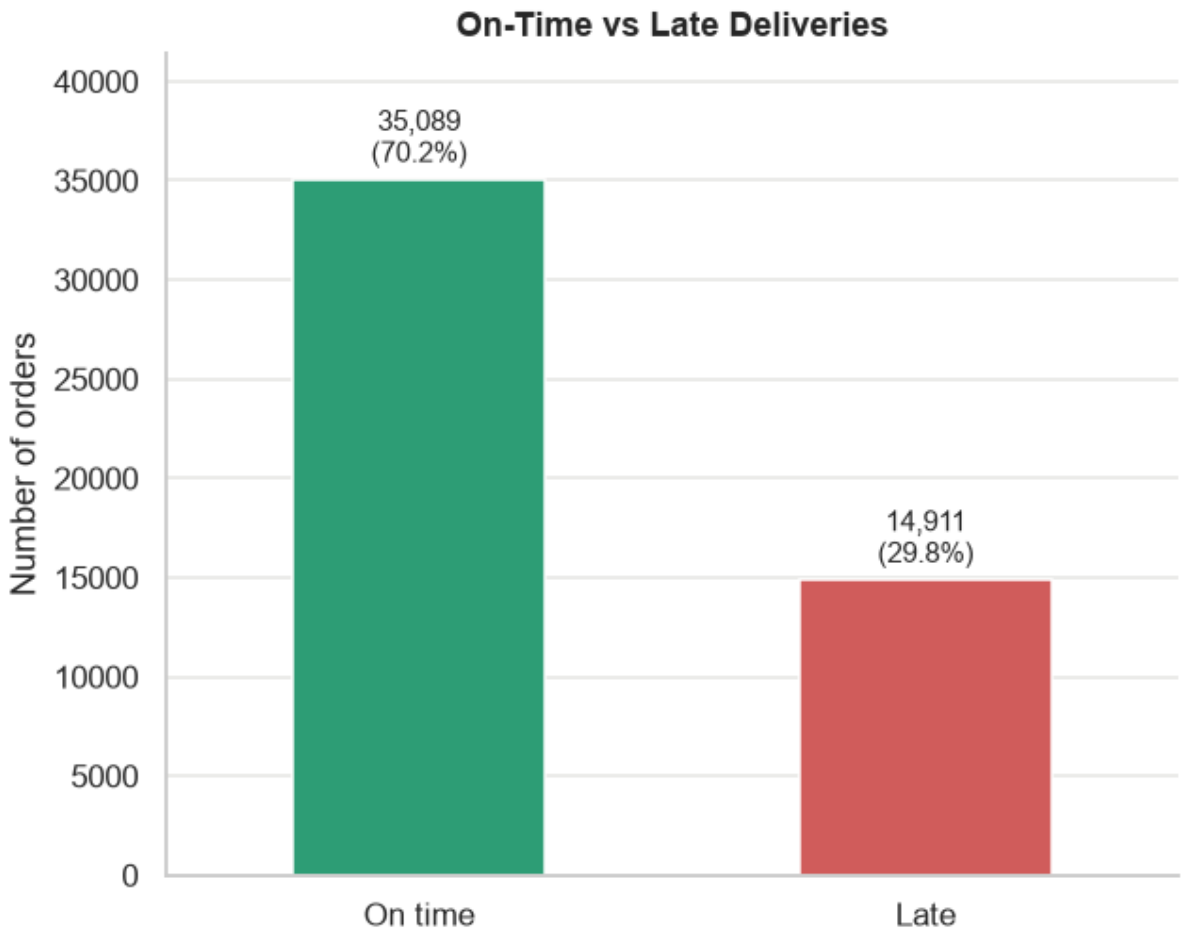
3. Charts and Findings

3.1 Delivery time distribution



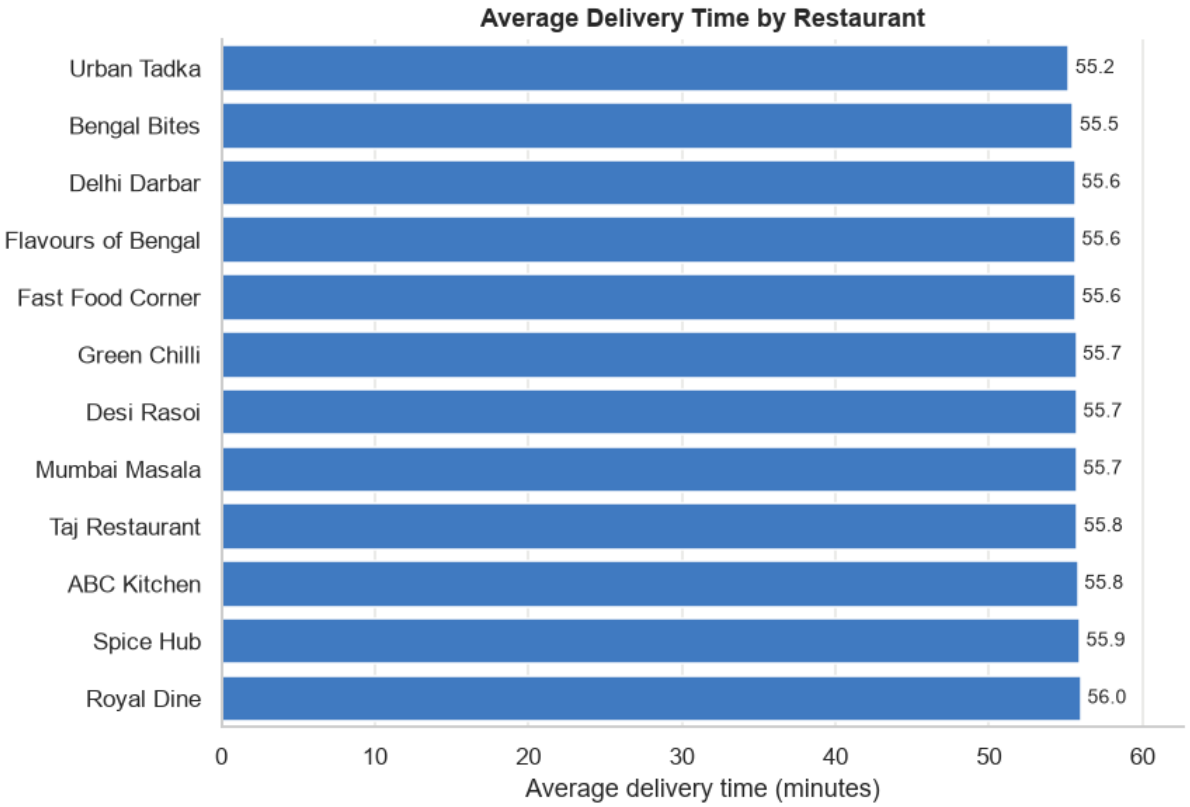
The average delivery time is 55.7 minutes. Most orders fall between 43 and 68 minutes (middle 50% of orders).

3.2 On-time vs late deliveries



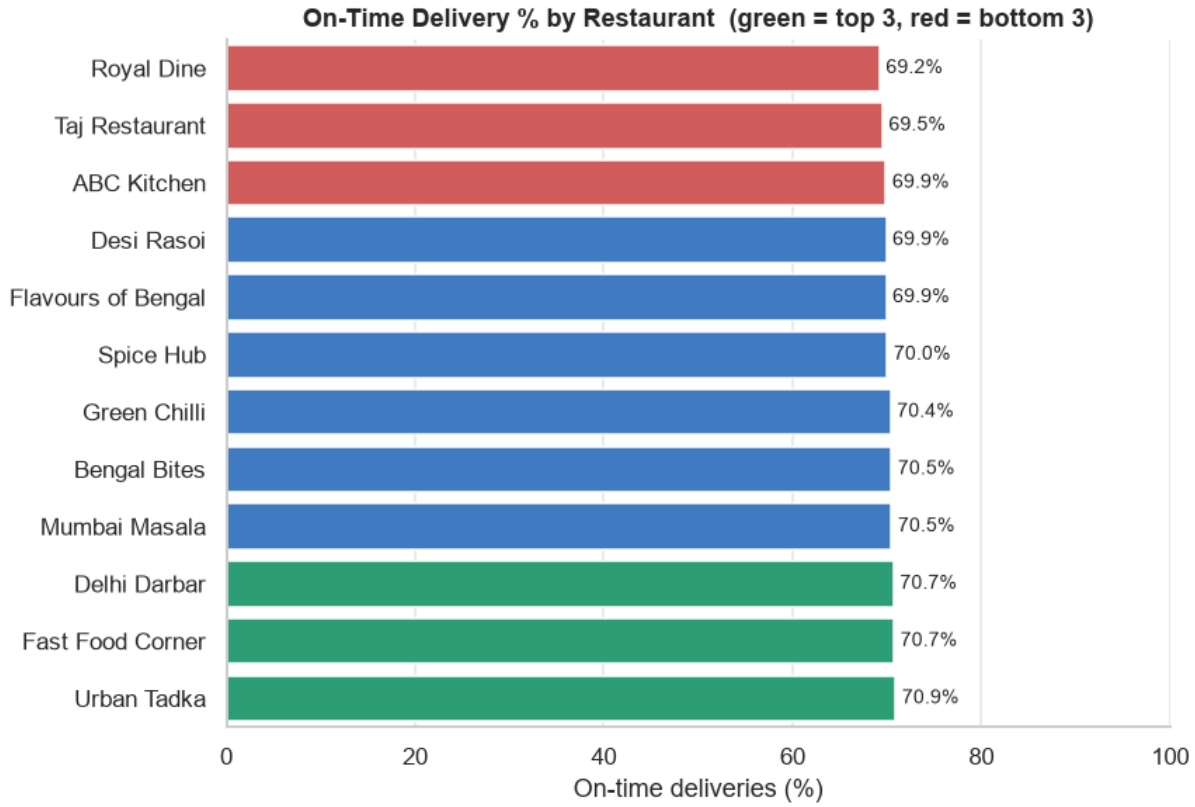
70.2% of orders were delivered within the expected time and 29.8% were late.

3.3 Average delivery time by restaurant



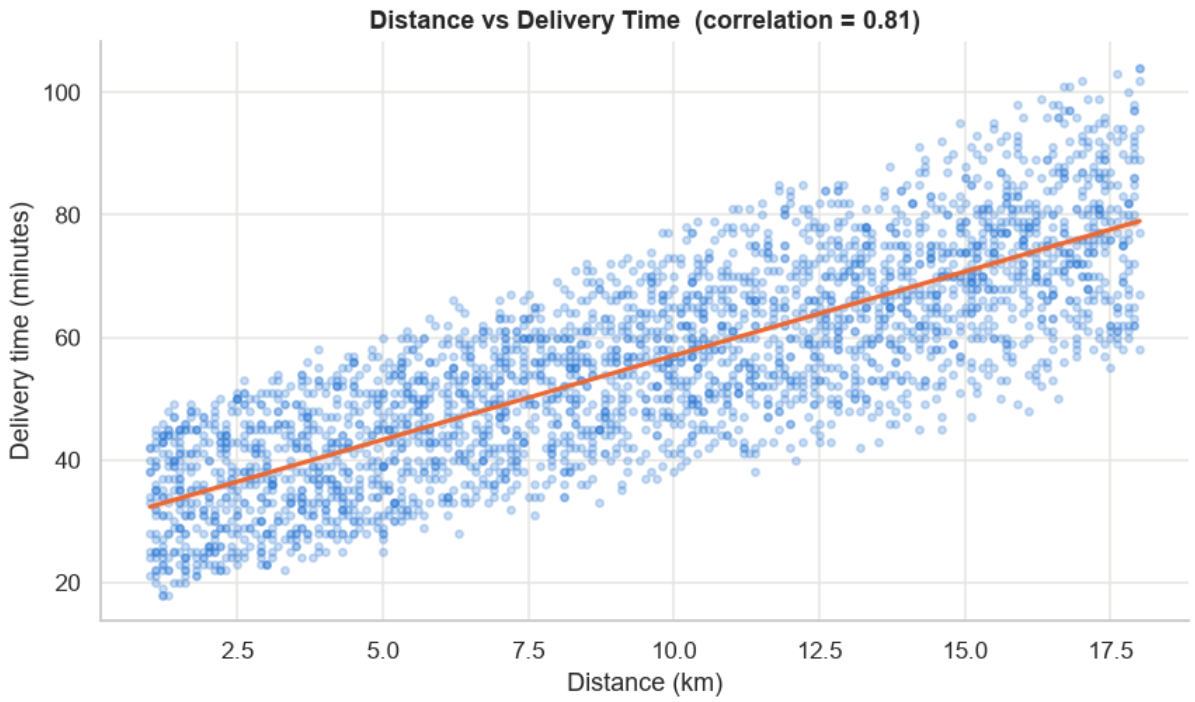
Restaurant averages range only from 55.2 to 56.0 minutes, so no restaurant is clearly faster than the others.

3.4 On-time delivery % by restaurant



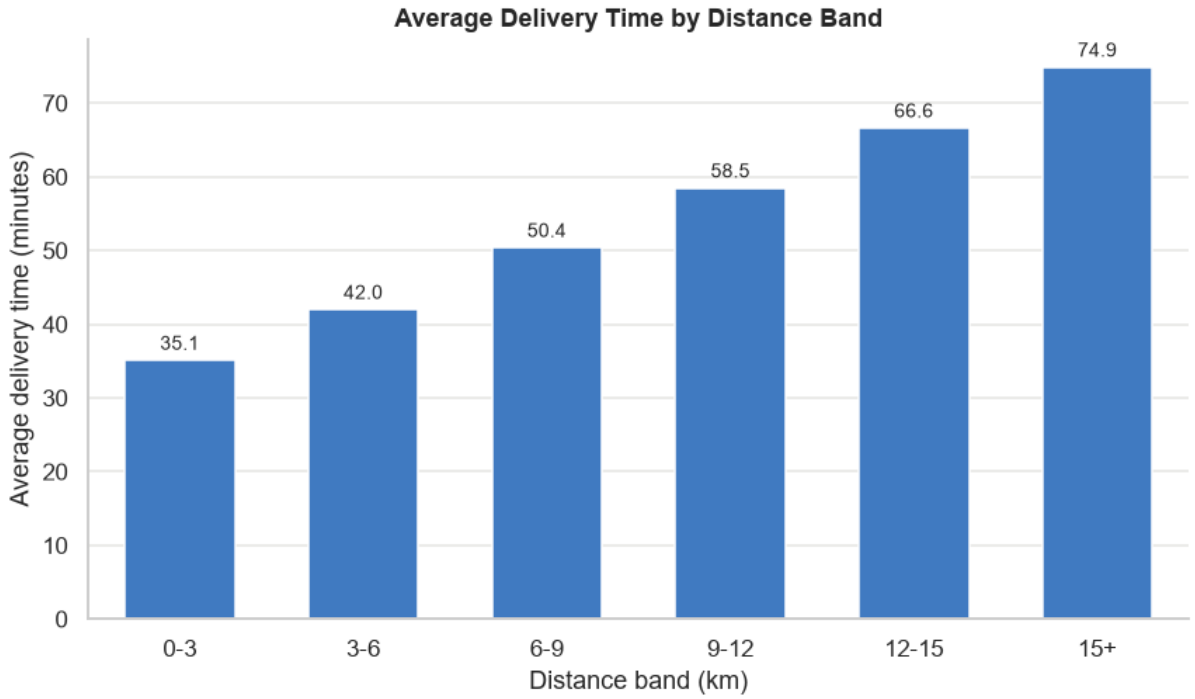
On-time rates range from 69.2% (Royal Dine) to 70.9% (Urban Tadka). The gap is only 1.7 percentage points, so the top/bottom ranking should not be treated as strong evidence of a real difference.

3.5 Distance vs delivery time



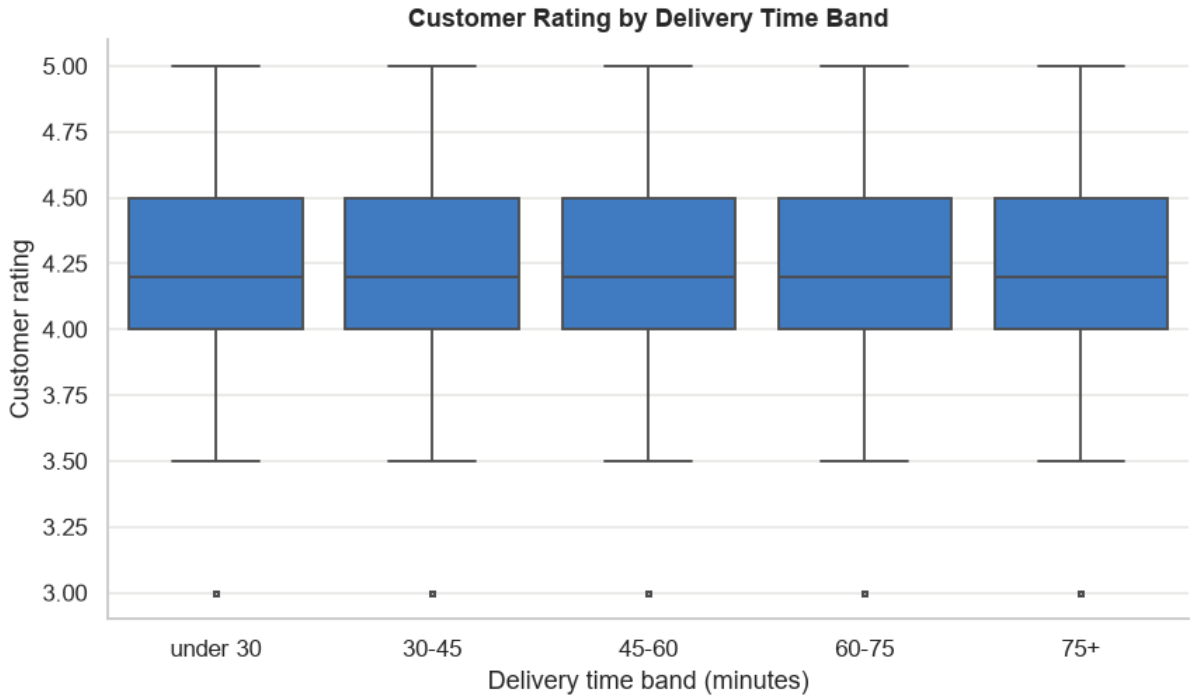
The correlation is 0.81: longer distance is strongly associated with longer delivery. Correlation does not prove cause.

3.6 Delivery time by distance band



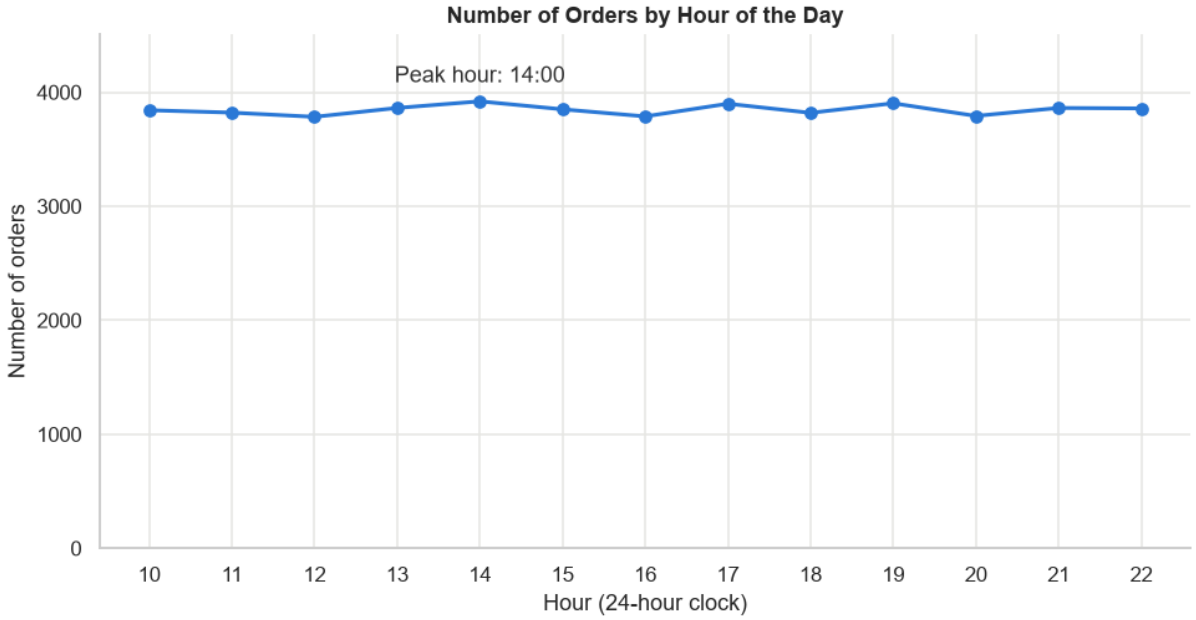
Orders within 3 km average 35.1 minutes, while orders over 15 km average 74.9 minutes.

3.7 Rating by delivery time band



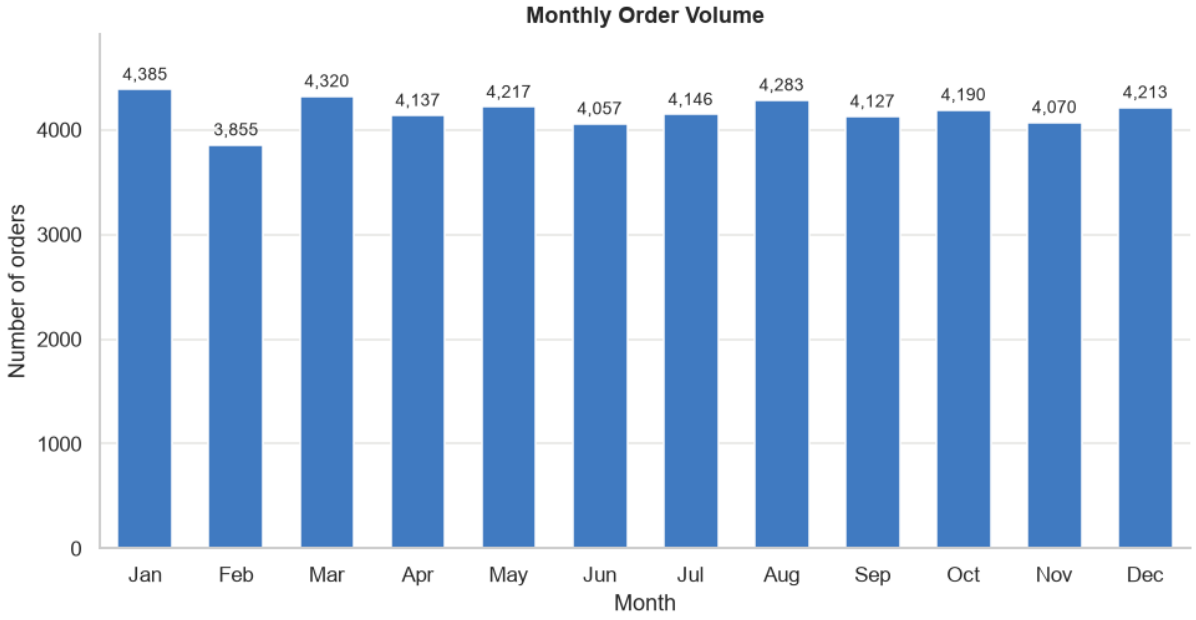
The correlation between delivery time and rating is 0.00, so in this dataset ratings do not change with delivery time.

3.8 Orders by hour of the day



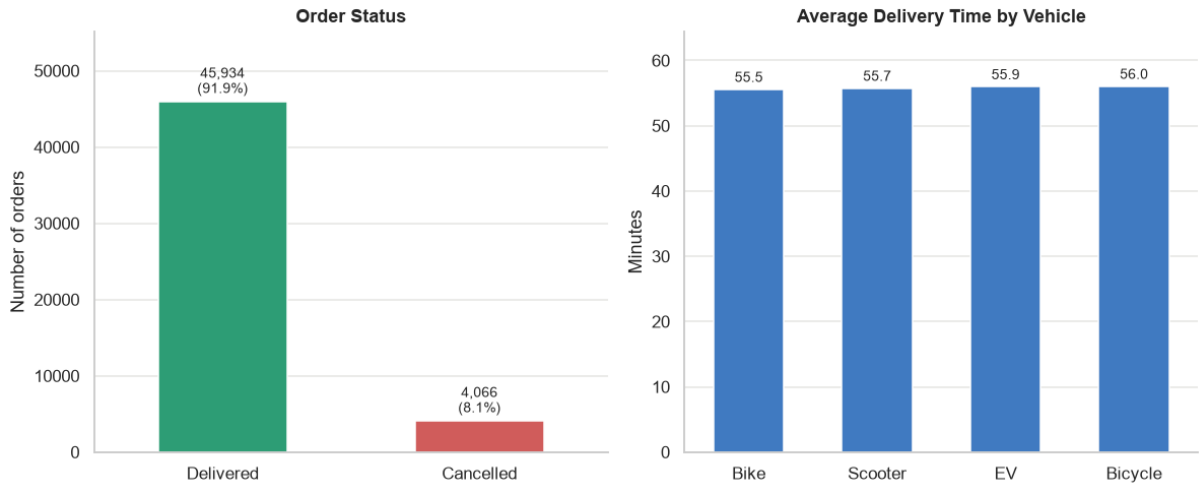
Order volume is highest at 14:00 (3,920 orders) and lowest at 12:00 (3,784 orders).

3.9 Monthly order volume



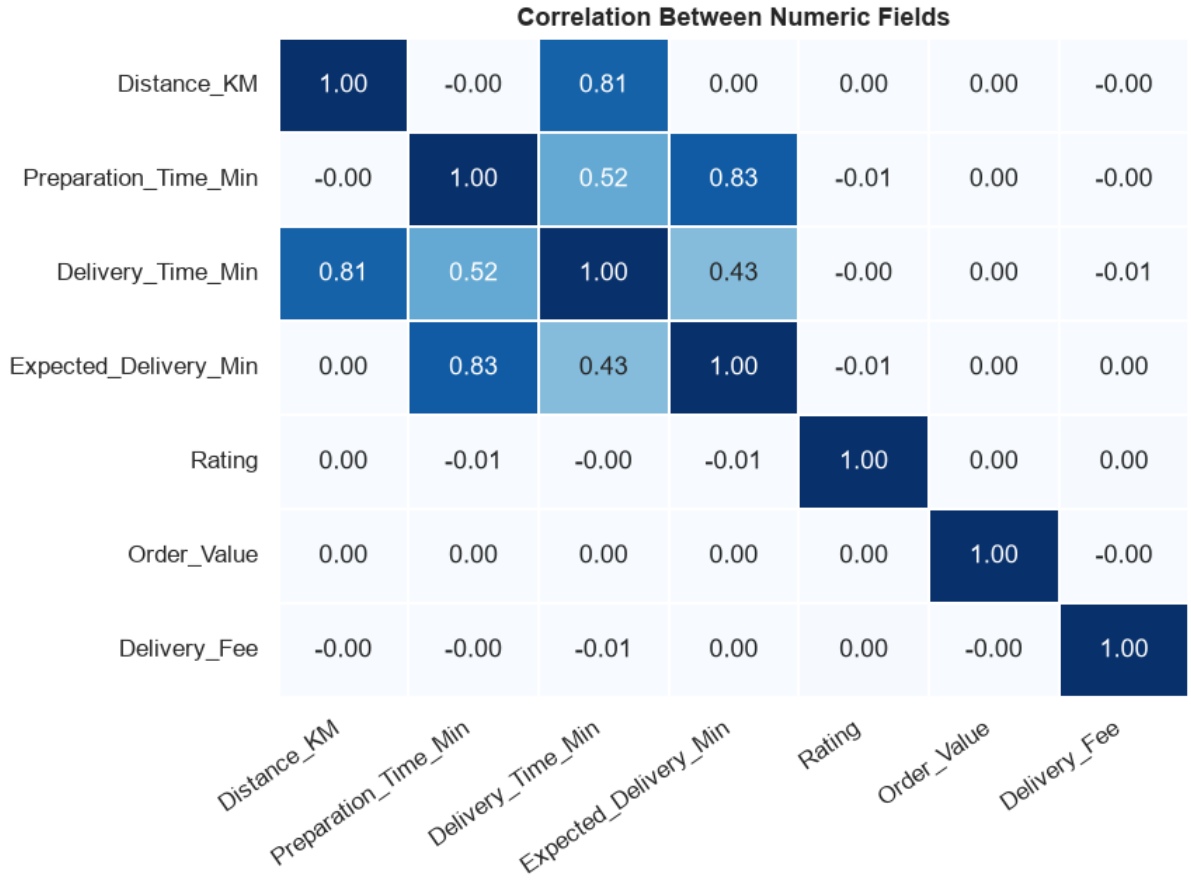
January has the most orders (4,385) and February the fewest (3,855).

3.10 Order status and vehicle type



8.1% of orders were cancelled. Average delivery time is almost the same for every vehicle type (55.5 to 56.0 minutes).

3.11 Correlation heatmap



Distance (0.81) and preparation time (0.52) are the two fields most related to delivery time. Rating, order value and delivery fee show almost no relation to it.

4. Summary of Findings

• Distance is the main driver of delivery time (correlation 0.81); preparation time is a smaller driver (0.52).

• About 70.2% of orders arrive on time, so roughly 30 in 100 orders are late.

• The restaurants perform very similarly: on-time rates differ by only 1.7 percentage points.

• Customer rating is not related to delivery time (correlation 0.00).

• Demand peaks at 14:00 and in January.

5. Limitations

• Charts show association, not cause.

• Restaurant differences are very small, so rankings may reflect chance.

• The dataset may be synthetic or simulated, so results may not match a real platform.

• Weather, traffic and rider availability are not in the data.

6. Power BI Dashboard

The interactive Power BI dashboard is built from the same cleaned CSV and the summary tables in *outputs/tables/*. Dashboard screenshots can be added here.